LocalMind

A Privacy-First AI Assistant for Document Analysis, Optimized for Snapdragon NPUs

EXECUTIVE SUMMARY

LocalMind is a fully offline, privacy-first AI document assistant engineered specifically to run locally on Snapdragon-powered HP PCs. By leveraging quantized open-source models optimized through the Qualcomm AI Hub, it empowers users to parse, query, and summarize confidential documents entirely on-device — without risking data leakage to external cloud environments.

TECHNICAL ARCHITECTURE & SNAPDRAGON OPTIMIZATION

- **NPU Acceleration:** Offloads model execution directly onto the Snapdragon Hexagon NPU to maximize throughput, minimize latency, and preserve battery life.
- **Quantized Models:** Leverages hardware-aware optimized models (e.g., INT4/INT8 variants) to fit comfortably within local device constraints while maintaining high accuracy.
- **Local Vector Storage:** Combines lightweight vector embedding databases with on-device retrieval-augmented generation (RAG) for instant document context retrieval.

APPLICATION USE CASE & INNOVATION

- Solves the critical enterprise and legal challenge of handling sensitive data by ensuring zero data transit across networks.
- Provides an intuitive workspace interface for legal professionals, researchers, and corporate workers to interact securely with local files (PDFs, Word docs).

DEPLOYMENT & ACCESSIBILITY

- Designed for seamless installation and local execution on Windows on Snapdragon devices.
- Includes complete open-source documentation, lightweight setup scripts, and clear instructions for running locally on HP Snapdragon hardware.